Study of Compton Scattering

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(Dated: March 13, 2026)

The Compton scattering of γ-rays from a Cs-137 source was studied using a calibrated NaI scintillation detector. Scattered photon spectra were recorded at different scattering angles for Aluminum and Brass targets. From the measured count rates and the Klein–Nishina differential cross-section, the calibration factors were obtained as $C_{\mathrm{Al}} = (3.61 \pm 0.03) \times 10^{30}\,\mathrm{sr\,m^{-2}\,s^{-1}}$ and $C_{\mathrm{Brass}} = (2.20 \pm 0.02) \times 10^{30}\,\mathrm{sr\,m^{-2}\,s^{-1}}$. The Compton wavelength was calculated using the angular dependence of wavelength of scattered photon to be $\lambda_C^{(\mathrm{Al})} = (0.0258 \pm 0.0004)\,\text{Å}$ and $\lambda_C^{(\mathrm{Brass})} = (0.0262 \pm 0.0006)\,\text{Å}$, which are close to the theoretical value. The results verify the angular dependence of wavelength shift predicted by Compton scattering.

I. OBJECTIVE

1. Energy calibration of scintillation detector
2. Determination of change in wavelength of the scattered gamma radiation as a function of the scattering angle
3. Determination of the differential cross-section using the Klein-Nishina formula and calculation of calibration factor.

II. THEORY

Compton scattering is a physical process involving the inelastic scattering of electromagnetic radiation (typically X-rays or gamma rays) by an electron. In 1920, Arthur Holly Compton observed that when X-rays were scattered from a carbon target, the scattered radiation had a longer wavelength than the incident radiation. This phenomenon provided critical evidence for the particle-like behaviour of light, demonstrating that photons carry both energy and momentum.

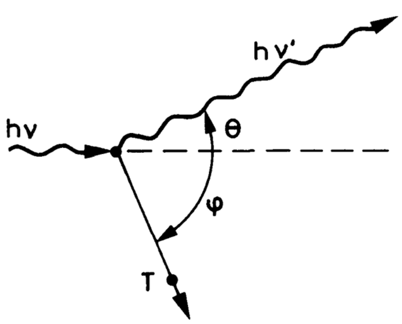

FIG. 1. Kinematics of Compton scattering (Adapted from [1])

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A. Kinematics of Compton Scattering

Figure 1 illustrates this scattering process. Applying energy and momentum conservation, the following relations can be obtained [1].

$$h\nu' = \frac{h\nu}{1+\gamma(1-\cos\theta)}, \tag{1}$$

$$T = h\nu - h\nu' = h\nu\,\frac{\gamma(1-\cos\theta)}{1+\gamma(1-\cos\theta)}, \tag{2}$$

$$\cos\theta = 1 - \frac{2}{(1+\gamma)^2\tan^2\phi + 1}, \tag{3}$$

$$\cot\phi = (1+\gamma)\tan\frac{\theta}{2}. \tag{4}$$

where $h\nu$ and $h\nu'$ are the energies of the incident and scattered photon, respectively. T is the kinetic energy of the recoil electron, θ is the scattering angle of the photon, ϕ is the recoil angle of the electron, $\gamma = \dfrac{h\nu}{m_e c^2}$ is a dimensionless parameter, h is Planck's constant, m_e is the rest mass of the electron, c is the speed of light,

The shift in the wavelength ($\Delta\lambda$) depends on the scattering angle θ. This relation is called the Compton formula [2]:

$$\lambda_\theta - \lambda_0 = \Delta\lambda = \frac{h}{m_e c}(1-\cos\theta) \tag{5}$$

where λ_0 is the wavelength of the incident photon, and λ_θ is the wavelength of the scattered photon.

The constant term $\dfrac{h}{m_e c}$ is known as the **Compton wavelength** of the electron and has a value of approximately 0.02426 Å.

For high-energy photons where $h\nu \gg 511$ keV, the scattered wavelength is typically on the order of the Compton wavelength. Conversely, in the low-energy regime ($h\nu \ll 511$ keV), the shift is minimal, and the behaviour approaches the classical limit, which is Thompson scattering.

B. Differential Cross-section

The cross-section for Compton scattering was one of the first to be calculated using quantum electrodynamics and is known as the **Klein-Nishina formula**. This formula accounts for quantum mechanical effects and is averaged over all incoming photon polarizations.

$$\frac{d\sigma}{d\Omega} = r_0^2 \left(\frac{1+\cos^2\theta}{2}\right)\left(\frac{1}{(1+\gamma(1-\cos\theta))^2}\right) \times \left[\frac{\gamma^2(1-\cos\theta)^2}{(1+\cos^2\theta)(1+\gamma(1-\cos\theta))}+1\right] \tag{6}$$

In this experiment, gamma rays from a Cs-137 source are used as the incident photons. The difference between the energies of the incident and scattered photons is measured using a calibrated scintillation detector positioned at different scattering angles. The relative intensities I_θ of the scattered radiation peaks are then compared with the predictions of the Klein–Nishina formula for the differential effective cross section $\left(\frac{d\sigma}{d\Omega}\right)$.

To make this comparison, a calibration factor C is calculated using the relation

$$C = \frac{1}{n}\sum_{\theta=0}^{n}\frac{I_\theta}{\left(\frac{d\sigma}{d\Omega}\right)} \tag{7}$$

III. APPARATUS REQUIRED

1. Cs-137 radioactive gamma source
2. Mixed preparation radioactive source for calibration (Am-241 and Cs-137)
3. Source holder in form of a lead block with a hole of 12 mm diameter at the centre to accommodate radioactive sources. An additional blind hole for inserting a steel pin as angular direction indicator
4. NaI scintillation detector and its holder with lead shielding for defined direction of incoming gamma radiation (refer to Figure 2)
5. High voltage power supply (1.5kV)
6. Cylindrical pure aluminium/copper rod as centre of scattering.
7. Additional lead shielding (movable) to reduce the intensity of unscattered gamma radiation, particularly for small scattering angles and short distances between source, scatterer and detector.
8. Multichannel analyzer (256 channels) and Related software in a desktop PC
9. Experimental panel with graduated angular scale

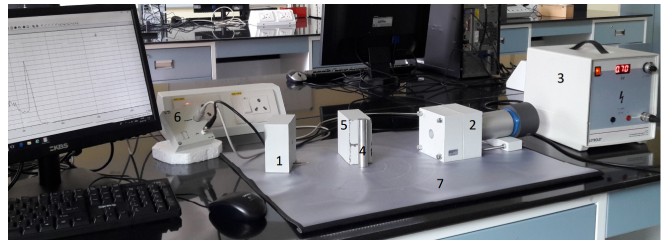

FIG. 2. Experimental set-up for Compton scattering experiment consisting of Source holder, Scintillator detector with lead shielding, High voltage supply, Scatterer, and additional movable shielding, MCA and an experimental panel with an angular scale (shown respectively as nos. 1-7)

IV. OBSERVATIONS

TABLE I. Energy calibration using the mixed source. (Day I)

S.no.	Source	Energy (keV)	Channel
1.	Am-241	59.54	21.79
2.	Cs-137	661.66	166.19

TABLE II. Fitting parameters for Aluminum scattering spectra. θ is the scattering angle, μ and σ are the mean and standard deviation of Gaussian fit to photopeak, Σ is the total counts under the photopeak, and I_θ is the intensity (count rate).

S.no.	θ (degree)	μ (keV)	Σ	σ (keV)	I_θ
1.	0	656.4	15323	18.3	153.23
2.	30	541.4	885	38.9	1.77
3.	45	462.3	702	33.3	1.404
4.	60	394.8	462	57.5	0.924
5.	90	281.2	401	19.4	0.802
6.	120	213.0	335	15.4	0.670

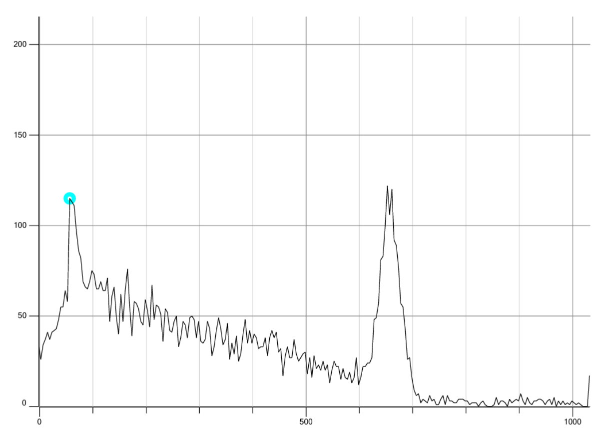

FIG. 3. Energy calibration spectrum obtained using the mixed radioactive source (Am-241 and Cs-137) (Day I)

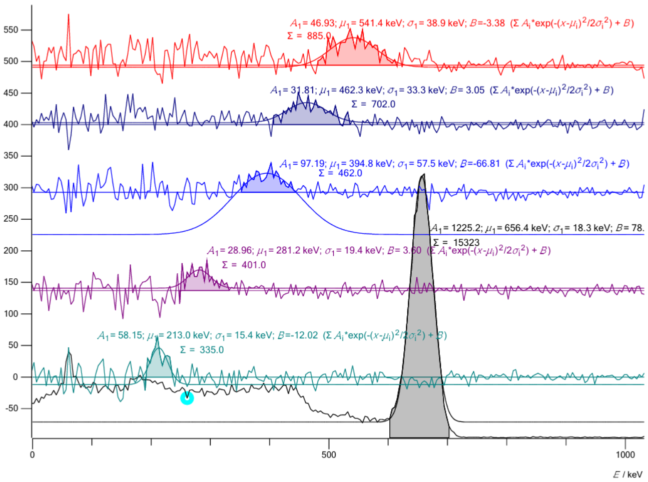

FIG. 4. Measured γ-ray spectra for the Aluminum scatterer at different scattering angles. Gaussian fits to the photopeaks were used to determine peak energy and intensity.

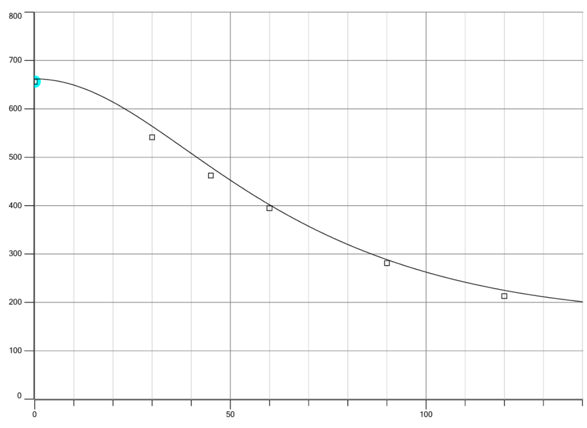

FIG. 5. Evaluation plot for Aluminum target showing the variation of scattered photon energy with scattering angle.

TABLE III. Energy calibration using the mixed source. (Day II)

S.no.	Source	Energy (keV)	Channel
1.	Am-241	59.54	20.943
2.	Cs-137	661.66	155.13

TABLE IV. Brass Scattering Spectra (fitting parameters)

S.no.	θ (degree)	μ (keV)	Σ	σ (keV)	I_θ
1.	0	655.8	7979	18.2	79.79
2.	30	548.1	1578	30.6	3.156
3.	45	467.2	1005	20.7	2.01
4.	60	395.8	803	37.8	1.606
5.	90	281.6	578	8.8	1.156
6.	120	210.6	199	51.8	0.398

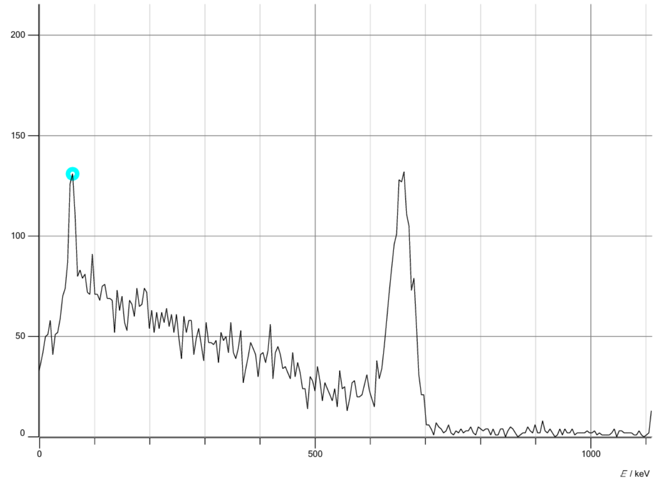

FIG. 6. Energy calibration spectrum obtained using the mixed radioactive source (Am-241 and Cs-137) (Day II)

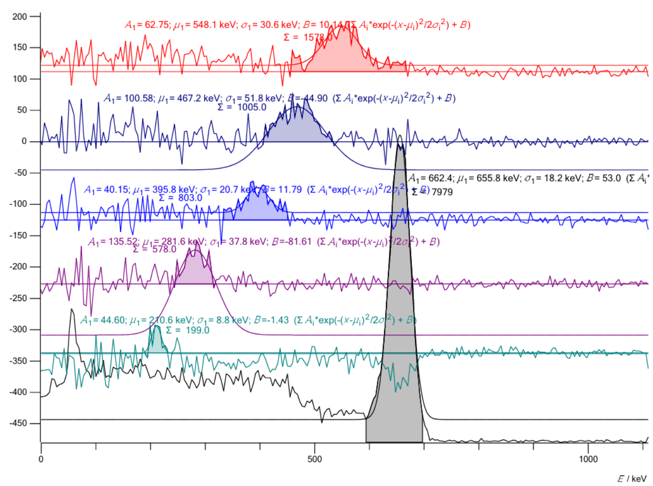

FIG. 7. Measured γ-ray spectra for the Brass scatterer at different scattering angles.

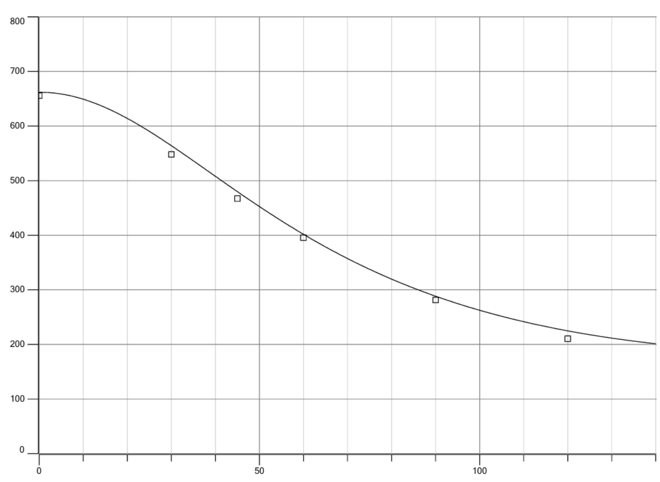

FIG. 8. Evaluation plot for Brass target showing the variation of scattered photon energy with scattering angle.

V. DATA ANALYSIS

A. Calibration Factor

The theoretical values of the differential cross-section $\left(\frac{d\sigma}{d\Omega}\right)$ calculated using the Klein–Nishina formula for different scattering angles are listed in Table V. The relative intensity I_θ were obtained from the area under the photopeak for the spectra. To account for different acquisition times, the peak areas were normalized by the measurement time, so that I_θ represents the count rate (s^{-1}) of the scattered radiation at angle θ. The count rates are listed in Table II and IV.

Angle θ (deg)	$\frac{d\sigma}{d\Omega}$ (m^2/sr) ($\times 10^{-30}$)
0	7.9411
30	5.1192
45	3.3477
60	2.2004
90	1.3045
120	1.1610

TABLE V. Differential cross section values for different scattering angles calculated using Klein-Nishina Formula (refer Eq. (6)).

The Poisson statistical nature of radioactive decay can be used to estimate uncertainty. The number of detected counts N in a time interval t follows Poisson statistics, giving $\Delta N = \sqrt{N}$. Since the intensities are expressed as count rates, $I = N/t$, to account for different acquisition times, the uncertainty in the rate is obtained by propagating the uncertainty in N. Propagating the uncertainty from N to the rate I (assuming no uncertainty in t) gives

$$\Delta I = \frac{\Delta N}{t} = \frac{\sqrt{N}}{t} \tag{8}$$

$$\implies \Delta I = \sqrt{\frac{I}{t}}. \tag{9}$$

The calibration factor is,

$$C = \frac{1}{n}\sum_{i=1}^{n} \frac{I_i}{\sigma_i},$$

where $\sigma_i \equiv \left(\frac{d\sigma}{d\Omega}\right)_i$ is the Klein–Nishina prediction for i^{th} angle. Considering no error in σ_i, the uncertainty in C becomes

$$\Delta C = \sqrt{\sum_{i=1}^{n}\left(\frac{\partial C}{\partial I_i}\Delta I_i\right)^2}.$$

Since $\frac{\partial C}{\partial I_i} = \frac{1}{n\sigma_i}$, the uncertainty in the calibration factor is

$$\Delta C = \frac{1}{n}\sqrt{\sum_{i=1}^{n}\frac{I_i}{\sigma_i^2\, t_i}}. \tag{10}$$

Using these values and the Eq. (7) and (10), we calculate the Calibration factor and its uncertainty to be,

For Aluminum Target,

$$C_{Al} = (3.61 \pm 0.03) \times 10^{30} sr\, m^{-2}\, s^{-1}$$

For Brass Target,

$$C_{Brass} = (2.20 \pm 0.02) \times 10^{30} sr\, m^{-2}\, s^{-1}$$

B. Compton Wavelength

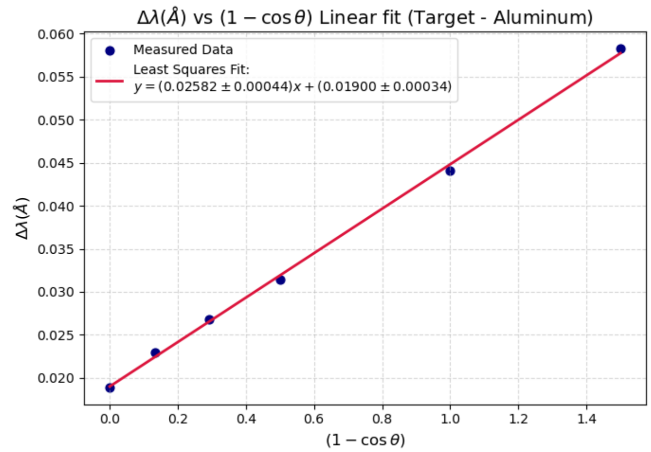

FIG. 9. Linear fit of the measured wavelength shift $\Delta\lambda$ versus $(1 - \cos\theta)$ used to determine the Compton wavelength using Aluminum target.

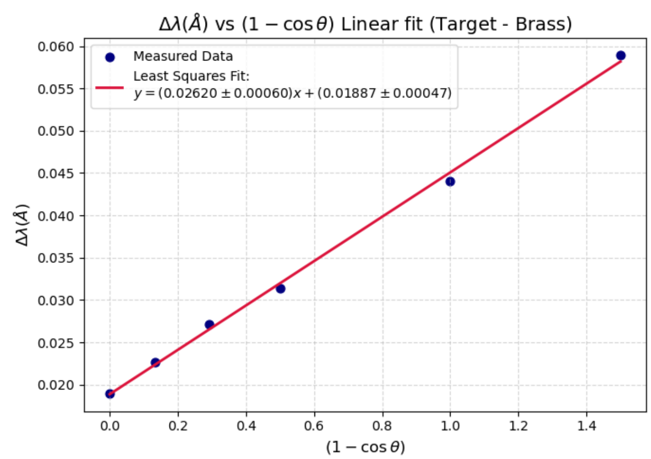

FIG. 10. Linear fit of $\Delta\lambda$ versus $(1-\cos\theta)$ for the brass target

To study the change in wavelength of scattered gamma radiation as a function of scattering angle, we perform a linear fit between $\Delta\lambda$ and $(1 - cos\theta)$, and obtain the Compton wavelength from the slope of the graph (see Figure 9 and 10).

We fit the observed data-points $\{x_i, y_i\}$ into a form,

$$y = mx + c$$

The statistical error calculated in linear fit is done using the following formulae [3],

$$\sigma_y = \sqrt{(\sum_i \frac{(y_n - y_i)^2}{N-2})} \tag{11}$$

Errors in slope is given by,

$$\Delta m = \sigma_y \sqrt{\frac{\sum_i x_i^2}{((N \sum_i x_i^2) - (\sum_i x_i)^2)}} \tag{12}$$

The slope and error in slope of the $\Delta\lambda$ vs. $(1 - \cos\theta)$ plot gives the Compton wavelength and its uncertainty:

For Aluminum target,

$$\lambda_{C_1} = (\text{slope}) = (0.0258 \pm 0.0004)\text{Å} \tag{13}$$

For Brass target,

$$\lambda_{C_2} = (\text{slope}) = (0.0262 \pm 0.0006)\text{Å} \tag{14}$$

The theoretical value of the Compton wavelength is $\lambda_C = 0.02426$ Å.

All the plots, data files and linear fit code are available on this GitHub Repository [4].

VI. RESULTS AND DISCUSSION

The scattered photon spectra were recorded for different scattering angles using the calibrated NaI scintillation detector. The photopeak energies obtained from Gaussian fits to the spectra decrease with increasing scattering angle, consistent with the expected behaviour for Compton scattering, as shown in Figure 5 and 8.

Using the measured count rates I_θ and the theoretical differential cross–section values from the Klein–Nishina formula, the calibration factors were calculated to be,

$$C_{\text{Al}} = (3.61 \pm 0.03) \times 10^{30}\,\text{sr m}^{-2}\text{s}^{-1}, \tag{15}$$

$$C_{\text{Brass}} = (2.20 \pm 0.02) \times 10^{30}\,\text{sr m}^{-2}\text{s}^{-1}. \tag{16}$$

A linear fit was performed between $\Delta\lambda$ and $(1 - \cos\theta)$ to verify the Compton formula and calculate Compton wavelength. It was calculated to be,

$$\lambda_C^{(\text{Al})} = (0.0258 \pm 0.0004)\,\text{Å}, \tag{17}$$

$$\lambda_C^{(\text{Brass})} = (0.0262 \pm 0.0006)\,\text{Å}. \tag{18}$$

These are close to the theoretical value of Compton wavelength, $\lambda_C = 0.02426$ Å. Minor deviations can be attributed to experimental uncertainties like finite detector resolution.

VII. CONCLUSION

In this experiment, Compton scattering of γ rays from a Cs-137 source was studied using a calibrated NaI scintillation detector. The angular dependence of the scattered photon wavelength was verified and used to determine the Compton wavelength through a linear fit of $\Delta\lambda$ versus $(1 - \cos\theta)$. The obtained values for aluminum and brass targets were found to be close to the theoretical Compton wavelength of the electron, showing reasonable agreement with the predictions of Compton scattering theory.

[1] William R. Leo. *Techniques for Nuclear and Particle Physics Experiments: A How-to Approach.* Springer, Berlin, Heidelberg, 2 edition, 1994.

[2] National Institute of Science Education and Research (NISER). *Study of Compton Scattering.* School of Physical Sciences, Bhubaneswar, India.

[3] John R. Taylor. *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements.* University Science Books, Sausalito, CA, 2nd edition, 1997.

[4] Aryan Shrivastava. P341 - nuclear physics and instrumentation lab. `https://github.com/crimsonpane23/P341-Nuclear-Physics-and-Instrumentation-Lab.git`, 2026.